FORM HS-08 · REV. 4 POST-STASIS REFRESHER · MODULE 8 OF 9 POST AT POLYMER WORKS AND EVERY CHEM STORE

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

READ THE LABEL.

THE POLYMERS, RESINS AND COATINGS THAT BUILD THIS CITY CAN BLIND YOU, BURN YOU OR RUIN YOUR LUNGS FOR LIFE.

SENSITIZED?

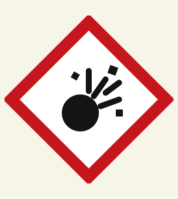
SOME CHEMICALS ONLY GIVE YOU ONE WARNING.

Isocyanates (spray foams, polyurethane coatings) and **epoxy resins** can make your body allergic to them. It can happen after one big exposure or after months of small ones. Once you are sensitized, **even a trace can set off an asthma attack or a skin reaction.** It does not go away.

Sensitized workers are moved off that material for good. Report wheezing, chest tightness, or an itchy rash on your hands or face to the Forward Aid Station the same shift.

THE 9 HAZARD SYMBOLS

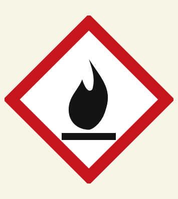
GHS PICTOGRAMS · ON EVERY CONTAINER



EXPLODING BOMB

Explosives, self-reactives, organic peroxides.

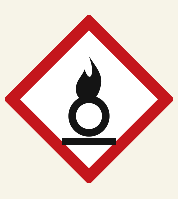
e.g. Blasting charges, some curing agents



FLAME

Flammable gas, liquid or solid. Can self-heat or ignite in air.

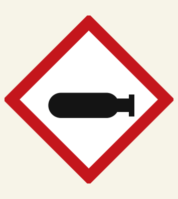
e.g. Solvents, fuel gas, VeilCoat



FLAME OVER CIRCLE

Oxidizer. Makes other things burn hotter and faster.

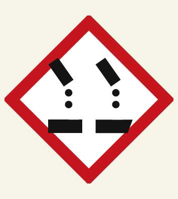
e.g. Oxygen cylinders, peroxides



GAS CYLINDER

Gas under pressure. Can explode if heated or damaged.

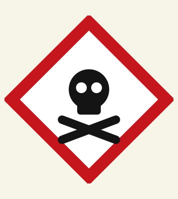
e.g. Welding gases, argon, nitrogen



CORROSION

Burns skin and eyes. Corrodes metal.

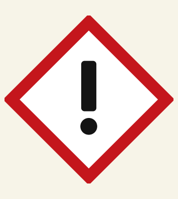
e.g. Dunecrete mix, acids, C-4X hardener



SKULL AND CROSSBONES

Toxic. Can kill in small amounts.

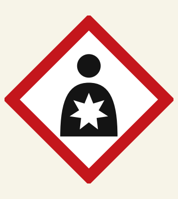
e.g. Cyanide compounds, some fumigants



EXCLAMATION MARK

Irritant, skin sensitizer, narcotic, harmful.

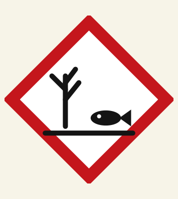
e.g. Most resins and cleaners



HEALTH HAZARD

Cancer, lung sensitizer, organ damage, reproductive harm.

e.g. Isocyanates, silica, hex chrome



ENVIRONMENT

Toxic to water life. Not required on site, still used.

e.g. Coolants, some coatings

Red diamond, black symbol, white background. If a container has no label, don't use it. Tag it and call Ch 15 Quartermaster.

DANGER

The more severe hazards. Can kill, blind, cause cancer, or burn on contact.

WARNING

Less severe, still harmful. Irritation, mild burns, harm with repeated exposure.

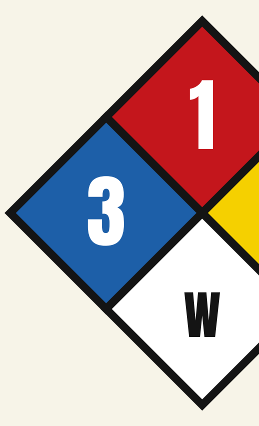
Every label carries one signal word, never both. The label also lists the hazard statements ("Causes severe skin burns") and what to do ("Wear protective gloves").

SAFETY DATA SHEET (SDS)

Every material on site has a 16-section SDS at the chem store and on the site terminal. The yellow ones are the sections you'll use on the job.

🟨 1 Identity & supplier	🟨 2 Hazards	3 Ingredients	🟨 4 First aid
🟨 5 Firefighting	6 Spill response	🟨 7 Handling & storage	🟨 8 Exposure limits & PPE
9 Physical properties	10 Stability & reactivity	🟨 11 Health effects	12 Ecology
13 Disposal	14 Transport	15 Regulations	16 Other

THE FIRE DIAMOND



- **Blue · Health**
- **Red · Flammability**
- **Yellow · Instability** (reactivity)
- **White · Special**: W = reacts with water, OX = oxidizer, SA = simple asphyxiant
- **0 = minimal, 4 = severe.** Example: QuickSet C-4X hardener (H3, F1, I1).

Fire & Rescue reads this diamond from 50 ft away. Keep it visible on every store and tank.

SITE MATERIALS

HAZARD CARDS · FULL SDS AT THE CHEM STORE

PRINTABLE STRUCTURAL POLYMER

FERROLATTICE-7

HEALTH HAZARD EXCLAMATION MARK

Extruded at about 500°F by the structure printers. Printing releases ultrafine particles and irritating fumes. The bead stays hot enough to burn for minutes after it looks set.

PPE: heat gloves, face shield at the head, half-mask with P100 + organic vapor cartridges inside print enclosures. Printer exhaust on before the head heats.

TWO-PART FAST-CURE COMPOSITE RESIN

QUICKSET C-4X

CORROSION HEALTH HAZARD EXCLAMATION MARK

Epoxy-isocyanate hybrid. Causes allergic skin rash and asthma. Mixing makes heat. **A large mixed batch can run away**, smoke, and ignite. Desert heat shortens pot life.

PPE: nitrile gloves (double), goggles, sleeves, respirator when spraying. Mix only the pot size on the label. Put leftover mix in the sand-filled cure bucket, spread thin, outside.

NANO-CERAMIC HEAT-SHIELD COATING

VEILCOAT

FLAME HEALTH HAZARD EXCLAMATION MARK

Sprayed on hulls and roofs to reflect sun. The solvent carrier is flammable. Spray mist carries nanoparticles deep into the lungs.

PPE: full-face respirator with P100 + organic vapor cartridges, coveralls, gloves. No hot work within 35 ft while spraying or until the solvent flashes off (see HS-05).

SAND-POLYMER BINDER (LOCAL SAND)

DUNECRETE

CORROSION HEALTH HAZARD

Strongly alkaline when wet, like cement. Causes chemical burns that start painless and get worse for hours. Dry mix and cutting cured Dunecrete make silica dust (see HS-02).

PPE: alkali-resistant gloves and boots, goggles, long sleeves. Wet cutting or vacuum shrouds only. Wash skin contact with water right away.

OLD EARTH HAZARDS THAT STILL APPLY

REAL CHEMISTRY BEHIND THE SITE MATERIALS

ISOCYANATES

In spray foams and polyurethane coatings. The leading cause of work-related asthma. Spray only with supplied air or the respirator on the permit. Keep bystanders out of the spray zone.

EPOXY RESINS

Uncured resin and hardener cause allergic dermatitis. It starts as a red, itchy rash on hands and wrists. Never wipe resin off skin with solvent. Use soap and water.

CURING HEAT

Resins give off heat as they cure. Mass speeds it up. A bucket of mixed resin can reach smoking temperature. Never leave mixed batches in a closed container or a trash bin.

STYRENE

Vapor from polyester and vinyl ester resins. Headache, dizziness, eye and throat irritation. Ventilate, and use organic vapor cartridges in tight spaces.

FIBER DUST

Cutting or sanding carbon fiber and fiberglass throws fine fibers and resin dust. They irritate skin, eyes and lungs. Wet cut or vacuum shroud, P100 respirator, sealed goggles.

BURNING POLYMERS

Polyurethane smoke releases **hydrogen cyanide**. PVC releases **hydrogen chloride**. All of them make **carbon monoxide**. Never fight a polymer fire without breathing air. Get out and call it in.

SPLASH

10 SECONDS TO WATER.
15 MINUTES UNDER IT.

Every eyewash and drench shower on site is within 10 seconds of the hazard, on the same level, with a clear path (ANSI Z358.1). Know where yours is before you open a container.

1. **Eyes:** get to the eyewash. Hold your eyelids open and roll your eyes. Flush **at least 15 minutes**. Take out contact lenses while you flush.
2. **Skin or clothing:** under the drench shower. Pull off contaminated clothing while the water runs. Flush 15 minutes.
3. **Breathed it in:** get to fresh air. Don't go back in for anyone without breathing air.
4. **Ch 1: "Emergency, emergency, emergency,"** with sector-grid, deck, the material, and how many are hurt. Then Ch 14 for the medic.
5. **Bring the SDS** or the container label to the medic. Section 4 tells them what to do.

STORAGE: KEEP APART

KEEP THIS	AWAY FROM	WHY
Oxidizers	Flammables, fuels, resins	Feed fire. Can ignite on contact.
Acids	Bases (alkalis), Dunecrete mix	Violent heat and spatter.
Acids	Cyanide or sulfide compounds	Releases toxic gas.
Water-reactives	Water, humid storage, sprinklers	Release flammable or toxic gas.
Isocyanate parts	Water, amines, heat	React, pressurize, burst drums.
Flammables	Sun, heat, sparks	Desert sun can push drums past flash point.

EVERY SHIFT

- **Check the label and the SDS** before you open anything new to you.
- **Wear the PPE in SDS section 8.** Not the PPE you happen to have.
- **Cap it when you're not pouring.** Open containers off-gas in the heat.
- **Never decant into an unlabeled container.** Secondary containers get a label too.
- **Wash hands** before eating, drinking or using the latrine. Resin on your hands ends up on your face.
- **Spills:** stop the source if it's safe, keep people back, call Ch 15 for the spill kit. Big or unknown spill: Ch 1.

UNKNOWN? DON'T TOUCH.

No label, a leaking drum, a smell you don't recognize, or a material you haven't been trained on: **stop work**, keep people back and call your foreman. No penalty, no questions.

WORKER ACKNOWLEDGEMENT

I have read this poster. I can read a GHS label and an SDS, I know where my nearest eyewash and drench shower are, and I will report any sign of sensitization.

NAME ____________________

COLONIST ID ____________________

CREW / SECTOR ____________________

DATE ____________________

Adapted from Old Earth standards: OSHA 29 CFR 1910.1200 Hazard Communication (UN GHS), NFPA 704 hazard identification, ANSI/ISEA Z358.1 emergency eyewash and shower equipment, NIOSH isocyanate and epoxy guidance, OSHA styrene and respiratory protection (1910.134) standards.